GIOVANNI QUINONES-VALDEZ

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RESEACH INTEREST

My research seeks to determine how genetic and post-transcriptional variants regulate the transcriptome, including gene expression, splicing, and polyadenylation, through the development of computational and statistical methods for multi-modal sequencing data. By enabling these analyses at scale, my work aims to decode the regulatory logic of RNA processing and its disruption in neurological disease, particularly Alzheimer's Disease.

EDUCATION

- **University of California, Los Angeles |** Bioengineering (Bioinformatics focus) | Ph.D. 12/2021
  Advisor: Dr. Xinshu (Grace) Xiao
  Dissertation: "Function and Regulation of Nucleotide Variants in RNA"
- **University of Florida |** Biological Engineering | B.S. 05/2015
  Minor: Biomolecular Engineering

EXPERIENCE

- **Postdoctoral Fellow |** *Department of Integrative Biology & Physiology, UCLA* 01/2022 – Present

  My research integrates statistical method development (isoLASER, scAllele) with multi-modal genomics data analysis (spanning long-read, single-cell, and bulk RNA sequencing) to study the role of RNA variants in basic transcriptomic regulation and disease.
- **Collaboratory Fellow |** *Institute for Quantitative & Computational Biosciences, The Collaboratory, UCLA* 01/2022 – Present

  I provided computational research support to UCLA laboratories through consulting and data analysis, and trained postdoctoral scientists and graduate students in bioinformatics through hands-on workshops.
- **Part-time Scientific Consultant |** *Rattan Life Science* 08/2025 – 09/2025

  Delivered hands-on training to staff scientists in the processing and analysis of single-cell RNA-sequencing (scRNA-seq) data analysis using R and artificial intelligence (AI) approaches for genomics data analyses.

PUBLICATIONS

1. Zheng R, **Quinones-Valdez G***, Xiao X*, Cheng C*. *CryEx.v2: An enhanced protocol for the quantitative detection and visualization of cryptic splicing using transcriptome data.* STAR Protocols. https://doi.org/10.1016/j.xpro.2026.104931. (2026) **(*Corresponding author)**
2. Barney RM, **Quinones-Valdez G**, King AJ, Amoah K, Wang W, Xiao X*. *Allele-specific alternative polyadenylation links noncoding genetic variation to Alzheimer's disease risk*. bioRxiv. https://doi.org/10.64898/2026.02.13.705798. (2026)
3. King AJ, Amoah K, Zhang L, Bahn JH, Barney RM, **Quinones-Valdez G**, Xiao X*. *Allele-specific splicing in the human brain reveals genetic variants linked to protein isoforms and Alzheimer's disease.* Genome Research. https://doi.org/10.1101/gr.281117.125. (2026)

4. **Quinones-Valdez G**, Chang JW, Magaki SD, Vinters HV, Salamon N, Liau L, Wang AC, Fallah A, Owens GC*. *Immune landscape of the affected brain in Rasmussen encephalitis.* Scientific Reports. https://doi.org/10.1038/s41598-026-51295-3. (2026).
5. **Quinones-Valdez G**, Amoah K, Xiao X*. *Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing*. Nature Communications. https://doi.org/10.1038/s41467-025-64605-6. (2025)
6. Hervoso JL, Amoah K, Dodson J, Choudhury M, Bhattacharya A, **Quinones-Valdez G**, Pasaniuc B*, Xiao X*. *Splicing-specific transcriptome-wide association uncovers genetic mechanisms for schizophrenia*. The American Journal of Human Genetics. https://doi.org/10.1016/j.ajhg.2024.06.001. (2024)
7. Zheng R, Dunlap M, Lyu J, Gonzalez-Figueroa C, Bobkov G, Harvey SE, Chan TC, **Quinones-Valdez G**, Choudhury M, Vuong A, Flynn RA, Chang HY, Xiao X*, Cheng C*. *LINE-associated cryptic splicing induces dsRNA-mediated interferon response and tumor immunity*. Molecular Cell. https://doi.org/10.1016/j.molcel.2024.05.004. (2024)
8. Liu Z, **Quinones-Valdez G**, Fu T, Choudhury M, Reese F, Mortazavi A, Xiao X*. *L-GIREMI uncovers RNA editing sites in long-read RNA-seq*. Genome Biology. https://doi.org/10.1186/s13059-023-03012-w. (2023)
9. **Quinones-Valdez G**, Fu T, Chan WT, Xiao X*. *scAllele: a versatile tool for the detection and analysis of variants in scRNA-seq.* Science Advances. https://doi.org/10.1126/sciadv.abn6398. (2022)
10. Chan TC, Fu T, Bahn JH, Jun HI, Lee JH, **Quinones-Valdez G**, Cheng C, Xiao X. *RNA editing in cancer impacts mRNA abundance in immune response pathways.* Genome Biology. https://doi.org/10.1186/s13059-020-02171-4. (2020)
11. **ENCODE Consortium.** *Perspectives on ENCODE*. Nature. https://doi.org/10.1038/s41586-020-2449-8. (2020)
12. **ENCODE Consortium**. *ENCODE Phase III: Building an Encyclopedia of Regulatory Elements for Human and Mouse*. Nature. https://doi.org/10.1038/s41586-020-2493-4. (2020)
13. **Quinones-Valdez G**, Tran S, Jun HI, Bahn JH, Yang EW, Zhan L, Brummer A, Wei X, VanNostrand E, Pratt G, Yeo GW, Graveley BR, Xiao X*. *Regulation of RNA editing by RNA Binding Proteins in Human Cells.* Communications Biology. https://doi.org/10.1038/s42003-018-0271-8. (2019)
14. Yang EW, Bahn JH, Hsiao YH, Tan BX, Sun Y, Fu T, Zhou B, Van Nostrand EL, Pratt, Freese P, Wei X, **Quinones-Valdez G**, Urban AE, Graveley BR, Burge CB, Yeo GW, Xiao X*. *Allele-specific binding of RNA Binding Proteins Reveal Functional Genetic Variants in the RNA*. Nature Communications. https://doi.org/10.1038/s41467-019-09292-w. (2019)
15. Hsiao YH, Bahn JH, Yang Y, Lin X, Tran S, Yang EW, **Quinones-Valdez G**, Xiao X*. *RNA editing in nascent RNA affects pre-mRNA splicing.* Genome Research. https://doi.org/10.1101/gr.231209.117. (2018)

ARTICLES UNDER REVIEW

1. Barney RM, **Quinones-Valdez G**, King AJ, Amoah K, Wang W, Xiao X*. *Allele-specific alternative polyadenylation links noncoding genetic variation to Alzheimer's disease risk*. Genome Biology.
2. Frenz-Wiessner S, Wang Y, Rohde Y, **Quinones-Valdez G**, Issa H, Mack P, Gonçalves-Dias J, Winkler R, Schuschel K, Ko MW, Wang Y, Wang Y, Ma F, Afshar Y, Sakamoto KM, Chang V, Xiao X, Klusmann JH*, Mikkola HKA*. *A persistent fetal-like progenitor underlies disease progression and relapse in Down syndrome myeloid leukemia*. Nature
3. Vavilina A, Calvanese V, Gu Y, Ma F, **Quinones-Valdez G**, Perrod C, Colombo G, Fares I, Ko MW, Kaaja I, Aguade-Gorgorio J, Nadel B, Liebscher S, Wang Y, Goodridge H, Crooks GM, Pellegrini M,

Xiao X, Schenke-Layland K, Mikkola HKA*. *Developmentally regulated long and short isoforms of MLLT3 balance human hematopoietic stem cell fate decisions.* Cell Stem Cell.

4. **ENCODE Consortium**. *The Encyclopedia of DNA Elements.* Nature

CONFERENCE ABSTRACTS

1. Kaaja I, Wang Y, Kilpivaara O, Wartiovaara-Kautto U, **Quinones-Valdez G**, Gu Y, Turmon A, Ma F, Wang Y, Aguade-Gorgorio J, Ko MW, Grimbert S, Wang Y, Katainen L, Park S, Pellegrini M, Xiao X, Seet CS, Chang VY, Arboleda V, Backus K, Mikkola HKA*. *Tracking the Stemness Mechanisms in Refractory Mecom Rearranged Acute Myeloid Leukemias at Single Cell Resolution*. European Hematology Association. 2025.
2. Biswal MR, Ildenfonso CJ, Rossmiller BP, Mao H, Li H, Zhu P, Tong Y. **Quinones-Valdez G**, Lewin AS*. *Delaying Retinal Degeneration in a Mouse Model of Geographic Atrophy: An Antioxidant Gene Therapy Approach*. Investigative Ophthalmology and Visual Science. 2015.
3. Biswal MR, Ildefonso CJ, Wang Z, Mao H, Li H, Zhu P, **Quinones-Valdez G**, Lewin AS*. *Effectiveness of Antioxidant Gene Therapy in Delaying Retinal Degeneration*. Molecular Therapy. 2015.

TEACHING AND MENTORING

- **Instructor for the workshop "Unix Command Line"** 01/2026 – 10/2026
  Institute for Quantitative and Computational Biosciences, UCLA (Quarterly)
- **Instructor for the workshop "Single Cell RNA-Seq. Data Analysis with R"** 01/2022 – 03/2025
  Institute for Quantitative and Computational Biosciences, UCLA (Quarterly)
- **Instructor for the Bruins in Genomics (BIG) summer program** 07/2024
  Institute for Quantitative and Computational Biosciences, UCLA 07/2023
  07/2020
- **Guest lecturer for the course "Intro to High-Throughput Data Analysis"** 03/2024 – 06/2024
  Department of Biomathematics, UCLA 03/2023 – 06/2023
- **Assistant instructor for the course "Concepts in Molecular Biosciences"** 01/2024 – 03/2024
  Department of Molecular Biology, UCLA
- **Teaching Assistant for the course "Molecular Parasitology"** 09/2017 – 12/2017
  Department of Microbiology, Immunology, and Molecular Genetics, UCLA
- **Member of the Xiao Lab at UCLA** 01/2016 – Present
  - Mentored junior graduate and undergraduate students.
  - Interviewed and hired students to aid with computational analysis.
  - Led 2 undergraduate students to first authorship on papers in *Genome Biology* and *Genome Research*

AWARDS

- **ASHG Trainee Research Excellence Award winner** 11/2024
  Awarded by the American Society of Human Genetics (ASHG) for outstanding trainee research presentation and scientific contribution at the annual meeting.

- **QCBio Collaboratory Postdoctoral Fellowship** 02/2022 – 03/2025
  Prestigious funding mechanism available to postdocs and project scientists conducting research in computational biology and bioinformatics.
- **Young Bolivian Abroad Award** 12/2018
  Recipient of the tarijeno abroad award (diaspora category), awarded by the regional government of Tarija, in recognition of academic achievement and contributions as a representative of Tarija abroad.
- **UCLA Bioengineering Supplementary Fellowship** 03/2017
  Fellowship support awarded by the UCLA department of bioengineering to support graduate research and training.
- **UCLA Bioengineering Research Fellowship** 01/2016
  Competitive graduate fellowship supporting research in bioengineering.
- **Davis Scholarship for Undergraduate Education** 08/2011 – 05/2015
  Merit-based scholarship awarded through the Davis United World College Scholars Program, supporting graduates of the United World Colleges network.

SELECTED RESEARCH TALKS

- **Title: "Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing"**
  - **American Society of Human Genetics, *Featured Plenary Talk*** 11/2024
  - UCLA Riboforum, *Selected Talk* 01/2024
  - International Society for Computational Biology, *Selected Talk* 11/2023
  - UCLA QCBio Retreat, *Selected Talk* 09/2023
- **Title: "scAllele, a versatile tool for the detection and analysis of variants scRNA-seq"**
  - Open Box Science, *Invited Talk* 04/2023
  - ENCODE Consortium Retreat, *Talk* 03/2022

JOURNAL REVIEWER

- Genome Biology 11/2025 – 10/2026
- BMC Genomics 01/2026
- Genomics, Proteomics, and Bioinformatics 08/2024
- Communications Biology 03/2023

SOFTWARE DEVELOPMENT

- **CryEx –** Lead Developer 10/2026

  Built in Python **|** *GitHub: https://github.com/giovanniquinones/CryEx, Pypi*

  CryEx is a computational pipeline for the detection and quantification of cryptic and annotated exons in RNA-seq data. CryEx reconstructs local isoforms from expressed splice junctions and de novo StringTie assemblies, enabling identification of rare, unannotated splicing events inaccessible to conventional tools.
- **isoLASER –** Lead Developer 10/2025

Built in Python | GitHub: https://github.com/gxiaolab/isoLASER, PyPi

isoLASER performs variant calling, haplotyping, and allele-specific splicing analysis in long-read RNA-Sequencing data. This one-stop solution enables the classification of genetically regulated splicing events revealing transcriptome-wide patterns of regulation and disease-relevant isoform regulation such as the HLA gene family and Alzheimer's disease risk genes.

- **scAllele –** Lead Developer 09/2022

  Built in Python | GitHub: https://github.com/gxiaolab/scAllele, PyPi

  scAllele is an integrative tool that performs high-confidence calls of small variants in single-cell RNA-Sequencing using a graph-theory approach. Additionally, scAllele enables allele-specific splicing analysis, a unique feature not previously used at the single-cell level.

- **L-GIREMI –** contributor 07/2023

  Built in Python | GitHub: https://github.com/gxiaolab/L-GIREMI

  L_GIREMI uses a robust statistical approach based on mutual information for the detection of A-to-I RNA editing in long-read RNA sequencing data unveiling insights about the distribution of these sites on individual RNA molecules.

- **BEAPR –** contributor 03/2019

  Built in C++, R | GitHub: https://github.com/gxiaolab/BEAPR

  BEAPR identifies allele-specific binding events in eCLIP-Seq data to functionally characterize genetic variants. Taking into account crosslinking-induced sequence propensity and variations between replicated experiments, BEAPR achieves high accuracy calls and identifies clinically relevant variants.

COMMUNITY ENGAGEMENT AND SERVICE

- **Diversity and Science Lectures (DASL)** 06/2024 – 08/2024

  Member of the DASL organization at UCLA that aims to promote diversity and inclusion through scientific seminars, social activities, and workshops.

- **Reviewer for the United World Colleges (UWC)** 10/2024

  Reviewed applications for the Bolivian UWC selection committee, contributing to the evaluation and selection of scholarship recipients. 10/2023

  10/2022

- **UCLA Latino Alumni Association** 10/2022 – 06/2023

  Guided undergraduate Latino students in strategizing their paths toward graduate school or securing employment opportunities.

- **Dashew Center for International Students and Scholars** 09/2016 – 06/2017

  Assisted incoming international students to help them adapt to the university and life in the United States

- **Engineering Graduate Student Association (eGSA)** 09/2015 – 12/2021
- **Health-Oriented Professional Engineering Students (HOPES)** 09/2014 – 05/2015
- **Society of Hispanic Professional Engineers (SHPE)** 09/2012 – 05/2015

SKILLS

- **Computational skills**

- Analysis of Next Generation Sequencing (NGS) data, including bulk RNA-seq, single-cell RNA-Seq, and Third Generation Sequencing data (PacBio, Nanopore)
- Software development using Python
- Software packaging and distribution (PyPI, Apptainer, GitHub)
- Machine Learning in Genomics using Scikit-Learn
- Programming languages: proficient with Python, Bash, and R
- Carried out projects on Google Cloud Platform (GCP)
- Operating Systems: Linux/Unix/Ubuntu, Mac OS, Windows

- **Languages**
  - Native or bilingual language proficiency: Spanish and English.
  - Limited working proficiency: French and Italian